

MinIO in Application Testing Environment (EC2 Setup)

1. Introduction MinIO is a high-performance object storage system compatible with Amazon S3 APIs. It is widely used in testing environments to simulate real cloud storage without incurring costs.

2. Why Use MinIO in Testing MinIO allows developers to safely test file uploads, backups, and logging without using production cloud storage. It reduces cost and avoids risk to real data.

3. Architecture Overview Application (EC2/Kubernetes) communicates with MinIO running on EC2. MinIO stores data on attached storage like EBS volumes.

4. Setup on EC2 Steps: - Launch EC2 instance - Install MinIO binary - Configure access key and secret key - Attach storage volume - Start MinIO service

5. Workflow - File Upload Step 1: Application sends request Step 2: Authentication Step 3: Bucket check Step 4: Store object Step 5: Response

6. Workflow - File Download Step 1: Request file Step 2: Retrieve from storage Step 3: Send to application

7. CI/CD Integration MinIO is used to store build artifacts, logs, and backup files during testing pipelines.

8. Benefits - Cost effective - S3 compatible - Easy to deploy - Fast performance - Safe testing

9. Best Practices - Use separate buckets - Enable versioning - Secure credentials - Monitor storage

10. Conclusion MinIO is an excellent tool for simulating cloud storage in testing environments.